

HANIF EDMA FAUZ

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EDUCATION LEVEL

Seoul National University of Science and Technology (SEOULTECH) Seoul, South Korea
Master of Electrical and Information Engineering, 4.19/4.50 Feb 2024 – Aug 2026 (Expected)

- Recipient of SEOULTECH scholarship; Research Assistant; specialization in control engineering and robotics

Universitas Indonesia Depok, Indonesia
Bachelor of Electrical Engineering, 3.61/4.00 Aug 2019 – Aug 2023

WORK EXPERIENCES

Control & Intelligent Robotics Lab SEOULTECH (CIRL SEOULTECH) Seoul, South Korea
Research Assistant Feb 2024 – Present

- **[Published, 1st Author]** “Performance Enhanced Risk-Aware MPPI using Gaussian Process for Robust Mobile Robot Control”, *IEEE Access* (SCI Q1, 2025); integrated offline Gaussian Process (GP) uncertainty estimation with Risk-Aware MPPI (RA-MPPI), validated on F1TENTH and Clearpath Jackal
- **[Submitted, 1st Author]** “Robust RA-MPPI using Online Learning”, submitted to *International Journal of Control, Automation and Systems* (IJCAS, Springer); online Sparse Gaussian Process (SGP) learning for real-time model correction and risk-aware trajectory sampling
- **[Ongoing]** Developing an imitation learning-based controller to replicate Model Predictive Contouring Control (MPCC) for mobile robot navigation, augmented with Disturbance Observer-Based (DOB) control
- **[Ongoing]** Developing a stability-guaranteed MPPI variant that samples feedback gain K instead of control input u , ensuring $u = -Kx$ satisfies local stability via closed-loop eigenvalue constraints

Universitas Indonesia Depok, Indonesia
Research Assistant (Undergraduate Thesis) Jan 2023 – Jun 2023

- Implemented an Equivalent Circuit Model (ECM)-based Adaptive Extended Kalman Filter (AEKF) with online parameter identification via Variable Recursive Least Squares (VRLS) for State of Charge (SoC) estimation of Li-Ion batteries in MATLAB simulation
- Achieved SoC RMSE of 0.02827 (LiFePO₄) and 0.01517 (LiCoO₂), outperforming Extended Kalman Filter (EKF) and Coulomb Counting baselines

DoctorTool (PT Medifa Infonyasa Suryantara) Jakarta, Indonesia
IoT Engineer Intern Feb 2022 – Jul 2022

- Developed an ESP32-based IoT gateway bridging multiple medical instruments (thermometers, oximeters, biometric sensors) to the cloud, supporting over six device categories

PERSONAL PROJECTS

Personal Website & Web Projects hanifedma.com
Full-Stack Developer

- Built and currently maintain a portfolio website (hanifedma.com) and two web apps: hanifedma.com/timbertimer and hanifedma.com/canopydiary

Corrosion Detection Application Jakarta, Indonesia
1st Champion – SciHackfest 2023 (Hackathon), PT Sucofindo Sep 2023

- Developed a computer vision (CV) system for corrosion detection and built the Android version of the application using Flutter (compatible with Windows, Linux, and Android) within a 1-month timeframe; awarded 1st place and a prize of 25 million IDR (~1,600 USD) as a team of 3 people

TECHNICAL SKILLS & LANGUAGES

- **Skills:** Electrical engineering, control engineering, IT, robotic engineering
- **Programming & Software:** MATLAB, Python, C, Robot Operating System (ROS1 and ROS2), Gazebo, Linux, Git, Docker, Web Development, Flutter, LaTeX, Microsoft Office
- **Hardware & IoT Devices:** Arduino, ESP32, Raspberry Pi, NVIDIA Jetson
- **Languages:** Bahasa Indonesia (native), English (IELTS Academic Band 7.0, 2023), Korean (beginner)